Exploration of Cloud Service Models (IaaS, PaaS, SaaS) Using a Free-Tier Cloud Platform

Case Study 6

Aim

To explore the three cloud service models – Infrastructure as a Service (IaaS), Platform as a Service (PaaS) and Software as a Service (SaaS) – using a free-tier cloud platform.

Objective

- Understand cloud computing and its service models.
- Create a free-tier cloud account and explore the console.
- Launch an IaaS resource (virtual machine) and a PaaS deployment.
- Use a SaaS application and compare responsibilities in each model.

Platform Used

Amazon Web Services (AWS) Free Tier. Similar free tiers are offered by Microsoft Azure and Google Cloud. Free-tier limits, credits and eligible services change over time, so the current terms on the provider's website should be checked before use.

Cloud Service Models

Model	What the provider manages	What the user manages	Example on AWS / Others
IaaS	Servers, storage, networking, virtualization	OS, runtime, middleware, application, data	Amazon EC2, Azure Virtual Machines
PaaS	Infrastructure + OS + runtime	Application and data	AWS Elastic Beanstalk, Google App Engine, Azure App Service
SaaS	Everything	Only usage and configuration	Gmail, Google Workspace, Microsoft 365, Zoho CRM

Responsibility Comparison

Layer	On-Premises	IaaS	PaaS	SaaS
Application	User	User	User	Provider
Data	User	User	User	Provider / User
Runtime & Middleware	User	User	Provider	Provider
Operating System	User	User	Provider	Provider
Virtualization, Servers, Storage, Network	User	Provider	Provider	Provider

Procedure

A. Account Setup

1 Visit the AWS Free Tier page and create an account (e-mail, password, contact and payment verification).

2 Sign in to the AWS Management Console and set a budget alert to avoid unexpected charges.

B. IaaS – Launch a Virtual Machine (EC2)

1. Open EC2 → Launch instance; choose an Amazon Linux AMI and a free-tier eligible instance type (e.g., t2.micro / t3.micro).
2. Create a key pair, allow SSH (port 22) and HTTP (port 80) in the security group, and launch.
3. Connect using SSH and install a web server:

```
sudo yum update -y
sudo yum install -y httpd
sudo systemctl start httpd
echo "Hello from IaaS" | sudo tee /var/www/html/index.html
```

1. Open the public IP address in a browser to view the page. Note that the user configured OS, patches and server software.

C. PaaS – Deploy an Application (Elastic Beanstalk)

1. Open Elastic Beanstalk → Create application; select a platform such as Python or Node.js.
2. Upload the sample application code or choose the sample application.
3. Environment (instances, load balancer, scaling) is created automatically; open the generated URL. The user only handled the code.

D. SaaS – Use a Ready-Made Application

Use Google Docs or Gmail through a browser: no installation, servers or code are required, and the provider handles updates and availability.

E. Clean-Up

Terminate the EC2 instance and delete the Elastic Beanstalk environment to avoid charges.

Observations

Parameter	IaaS (EC2)	PaaS (Beanstalk)	SaaS (Gmail / Docs)
Setup effort	High	Medium	None
Control	Full control of OS	Control of code	Minimal
Maintenance by user	OS patches, security, scaling	Application only	None
Scalability	Manual / configured	Automatic	Automatic
Best for	Admins, custom setups	Developers	End users

Results

- A virtual machine (IaaS) and a web application (PaaS) were deployed on a free-tier account, and a SaaS tool was used.
- The level of user control decreases from IaaS to PaaS to SaaS, while convenience increases.
- Cleaning up resources is essential to stay within free-tier limits.

Conclusion

Each cloud service model suits a different need: IaaS offers flexibility, PaaS speeds up application development, and SaaS delivers ready-to-use software. Free-tier platforms are a practical, low-cost way to learn these models hands-on.